A REPORT

ON

AI CLIMATE RISK INTELLIGECE PLATFORM

Submitted by

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Under the guidance of,

Ms. Shet Reshma Prakash
Mr. Sachin Kumar

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY
(COMPUTER ENGINEERING)

AT

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **"AI CLIMATE RISK INTELLIGENCE PLATFORM"** being submitted by "**MOHAMED FAHAD**" bearing roll number "**20241CSG0033**" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Technology (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**AI CLIMATE RISK INTELLIGENCE PLATFORM"** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Technology (Computer Science and Engineering),** is a record of my own investigations carried under the guidance of **Ms. Shet Reshma Prakash,** Professor and **Mr. Sachin Kumar,** Professor Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

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ACKNOWLEDGEMENTS

First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

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ABSTRACT

AI Climate Risk Intelligence Platform: A Full-Stack, Hardware-Accelerated Machine Learning System for Meteorological Hazard Assessment

Abstract:
With the escalating frequency of extreme meteorological events, scalable and precise risk assessment systems are critical. This paper presents the AI Climate Risk Intelligence Platform, a full-stack system that ingests over 530,000 meteorological records, cleans and scales the features, and trains regression models to predict regional hazard indexes. Among candidate models, the Random Forest Regressor demonstrated superior performance, achieving an R^2 score of 89.27%, an RMSE of 0.3285, and an MAE of 0.2363.

To serve these analytics, we built an enterprise web dashboard containing an operator console, ML pipeline visualizers (correlation heatmaps and feature importances), a 5-day weather predictor, and a paginated dataset explorer. The system integrates NVIDIA H200 GPU acceleration via CUDA to minimize prediction latency (<150ms). Security is managed using JWT role-based access control and administrative audit logging, while PWA caching guarantees offline usability. Ultimately, the platform bridges the gap between raw meteorological datasets and actionable, real-time climate hazard intelligence.

Keywords: Climate Risk Assessment, Machine Learning, Random Forest Regressor, CUDA GPU Acceleration, Full-Stack Architecture.

1. INTRODUCTION:

Climate change and weather volatility create critical risks for industries and communities. Traditional systems rely on static historical averages and fail to capture real-time weather fluctuations. The AI Climate Risk Intelligence Platform is a full-stack client-server application built with Node.js and Python to calculate and visualize localized risk indices. It features interactive Chart.js visualizations, offline PWA access, and low-latency predictions using host GPU acceleration.

2. OBJECTIVES OF THE WORK

- Ingest and clean a dataset of 530,000+ raw climate records (Combined12.csv) using IQR outlier removal.
- Train and evaluate multiple ML models to predict localized environmental risk indices.
- Implement a multi-output forecast engine to project weather vectors over a 5-day horizon.
- Build a secure dashboard console using role-based JWT session controls and user audit logs.
- Integrate host hardware probes to utilize GPU acceleration (NVIDIA H200) for real-time model inference.

3. WORK PROGRESS & SYSTEM IMPLEMENTATION:

Data engineering cleaned 533,312 records using column average imputation and Interquartile Range (IQR) outlier capping. A regression pipeline trained Linear Regression, Random Forest, and Gradient Boosting models, saving the highest-performing Random Forest model (best_model.pkl) for server scoring. A 5-day forecast model was built using autoregressive lags. The Express.js backend serves JWT-secured prediction APIs, SQLite audit logging, and PDFKit report generation, utilizing CUDA GPU acceleration detected via nvidia-smi.

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4. RESULTS & SYSTEM VERIFICATION:

The Random Forest model achieved a high accuracy of 89.27% R2, outperforming Gradient Boosting (86.14% R2) and Linear Regression (59.45% R2). Feature importance weights showed that temperature profiles (temp_max and temp_mean) contribute approximately 75% of the risk scoring weight. End-to-end verification confirmed successful 5-day forecasting, JWT security, data explorer pagination, and automated PDF assessment reports running on active GPUs.

5. CONCLUSION:

The internship successfully delivered a full-stack, AI-powered climate risk platform optimized for GPU-accelerated computing nodes. By implementing automated ML pipelines, role-based JWT security, 5-day forecasting, and automated PDF reporting, the platform meets all key engineering objectives. This project provided valuable practical experience in full-stack engineering, machine learning pipelines, and hardware integration.

6. REFERENCES:

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